Phase 3C | First-coach assembly development

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

A coordinated path to the first coach

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Preliminary engineering study - not for fabrication, construction, procurement release or passenger operation. This issue advances the regular-coach chassis and assembly interfaces. Accessible-coach changes are identified separately; a common outer envelope does not establish an interchangeable structure.

Retain the owner's **0% longitudinal grade** target, 36-inch evaluation gauge, three regular coaches and one separate accessible coach, diesel propulsion, enclosed conditioned cabins and hydraulic service-brake preference. North Pole location remains deferred; passenger disembarkation and accessible boarding are required at both stops.

Work developed in this issue	Evidence still required
A coordinate-based frame grid, actual member segments and comparative load/weight calculations.	Engineer-selected load spectrum, members, joints, fatigue and stability criteria; exact bogie reactions and movement envelope.
Coordinated floor/wall/roof stacks, supported panel seams and accessible boarding alternatives.	Occupied mockup results, manufacturer application approval, panel/seat/door attachments and verified accessible evacuation.
Functional brake/control interfaces, failure responses, connection load inputs and measurement records.	Configured OEM hardware, accepted circuit design, electrical/hydraulic ratings, installed timing and fault-test acceptance limits.
Fabrication sequence, datum scheme, inspection traveler and first-article handoffs.	Released shop drawings, adopted contractual welding requirements, qualified fabrication and responsible inspection.

The CAD describes selected-for-study geometry and explicitly held interfaces. It is not a substitute for finished shop drawings. No supplier-approved project ratings, occupied-mockup results or professional engineering approval were received or assumed in producing this issue.

01 | What changed and what remains controlled

Disposition	Basis / decision
Retained	Phase3B occupancy, 306 x104 in clear cabin, 312 x110 in outer body, 400 in frame study length and x45/261 in bogie centres. The 28 in finished floor is still a target subject to real bogie and structural packaging.
Retained	Original 7,040 kg regular /7,535 kg accessible tare allocations; 54,755 kg nominal and59,840 kg stress train budgets. Upper calculation cases remain13,200 /11,000 kg per car and65,100 kg complete train.
Revised in this study	Model actual member segments at continuous girders instead of overlapping solid tubes; reconcile floor finish, panel, isolation and support planes. Count additional steel and body components openly rather than treating early allocations as proven masses.
Developed further	Supported panel seams, body/utility access, functional hydraulic controls, loss-of-power/connection responses, jig datums, assembly dependencies and measurement records.
Not accepted as a construction basis	A catalogue gauge match, whole-train brochure rating, generic brake force or attractive CAD view does not establish this coach's compatibility. No missing weld, pin, bolt, adhesive or test limit is silently filled in.
Deferred without blocking vehicle development	Surveyed final alignment, North Pole position and site-specific stations. The owner's flat-track target does not remove drainage, crosslevel, twist, settlement or as-built acceptance work.

The retained 32-seat regular layout is the first mockup to test, not a frozen production interior. The Phase3B alternatives remain available. Changes in seating, floor height, body mass or loading pattern must be carried through bogie loads, stability, brakes, couplings, HVAC and stations.

02 | Coordinate system and interface ownership

Controlled item	Study basis / owner
Coordinate datum	Inches: x=front inside end wall toward rear; y=transverse centreline; z=top of rail. The Phase3B occupied mockup uses its own ground-floor datum and must not be placed at rail level by mistake.
Vehicle geometry	Frame x=-47 to353; body x=-3 to309; clear cabin x=0 to306. Side envelopes y=+/-55 outer /+/-52 clear. Bogie centres x45/261. These are engineering study inputs, not production tolerances.
Structure to running gear	Vehicle engineer and bogie supplier jointly establish sprung/unsprung masses, pivot/sidebearer forces, roll path, attachment, retention, suspension travel and full movement clearances.
Structure to body	Floor, wall, roof, glazing, seats, guards, stairs, lift/bridge and utilities deliver discrete loads into reviewed frame nodes. Finishes and adhesives are not credited as structural without supporting qualification.
Brakes and train control	Brake integrator owns configured torque, stored energy, safe state, isolation, timing and feedback. Vehicle engineer owns frame/axle reaction paths; locomotive supplier owns compatible command and traction interlocks.
Track and stations	Track engineer owns matched rail/wheel geometry, alignment/tolerances and test route. Station/accessibility designers own boarding gaps, bridge/ramp limits, passenger protection and emergency interfaces at both stops.

Every unresolved interface needs one responsible person, an exact required return, a due point in the sequence and a documented acceptance. The workbook provides those fields; an empty field means open, not approved.

03 | Chassis grid, load paths and member studies

The first regular coach now has an explicit frame grid, seat-to-frame load path, floor-support comparison and disjoint material quantity model. This is a benchmark for engineering optimization. The modeled bare members weigh **1,913 kg**, already **413 kg above the retained 1,500 kg underframe allocation**, before connection reinforcement and several necessary assemblies. Therefore the earlier allowance is not demonstrated; neither the 7,040 kg coach tare target nor the 13,200 kg upper calculation case has been silently changed. The useful next decision is a supplier-supported floor span and bogie interface, followed by complete frame/joint analysis and a revised mass allocation.

1. Geometry that can be coordinated now

Use inches: x0 at the front inside end wall, y0 on the vehicle centreline, z0 at top of rail. Retain a 400-inch frame from x−47 to353; body x−3 to309, 110 inches outside and104 inches clear; bogie centres x45/261; 36-inch evaluation gauge and **0% longitudinal grade**. The body is not credited as a structural truss.

The primary study uses two continuous HSS12×4×¼ girders at y±34. Transverse members terminate at their faces rather than occupying the same solid volume. Every transverse node has three study segments: y−53…−36, −32…32 and36…53, totaling 98 inches. At x45/261, HSS12×4×¼ bolster segments replace ordinary crossmembers. This geometric split does **not** establish welds, moment continuity or local HSS capacity; all these joints are held for design.

Ordinary crossmembers are HSS4×2×3/16, four inches vertically and two inches longitudinally. Cabin node stations are **−3,13,29,45,61,77,93,109,125,141,157,173,189,205,221,237,253,261,269,285,301,309**. The maximum centre spacing is 16 inches, with shortened bays near the rear bolster/end. Platform nodes are x−46,−25,331,352. Perimeter rails are HSS4×2×3/16 at y±54. The 110-inch width remains an envelope, not a track clearance approval.

Four HSS3×1×⅛ longitudinal seat-attachment rail lines at y−44,−20,20,44 are segmented between transverse-member faces. They provide a defined route for seat feet into steel without assuming honeycomb skins can carry anchor loads. Their locations and section are study choices; finished-seat supplier bases, fastening and fatigue requirements remain open. They do not reduce the 104-inch cabin width.

The body study's proposed110-inch-wide floor-panel seams at x45,93,141,189,237,285 all land on this grid. The former three-strip48-inch-sheet nest had longitudinal seams away from support rails. The revised seam scheme is coordinated, but panel bearing width, closures, screws, inserts and adhesive details remain unapproved.

2. The 28-inch floor still has an unresolved bogie fit

The previous panel-only support plane is superseded. The explicit packaging trial is2.25mm Gerflor NT finish, **1mm unselected finish-adhesive allowance**,19.05mm panel, and **3mm unselected isolation/bedding allowance**. Total25.30mm places the steel bearing plane at **z27.00394 inches**, and a12-inch-deep member underside at **z15.00394 inches**. The 10-inch girder alternative ends atz17.00394, but the retained 12-inch bolsters still extend toz15.00394. Changing girders alone does not raise the bolster interface.

Gerflor's NT and MK thicknesses are documented C3S03. The adhesive/bedding allowances are geometric inputs, not installation prescriptions. MK finish would lower the support plane another 0.15551 inches. Body insulation can occupy bays only after thermal, fire, water and equipment-clearance review.

At the centre-bolster envelope x43…47 and259…263, y−32…32, the condition is:

bogie support-top height + attachment stack + required relative-motion clearance ≤15.00394 inches above rail.

With assumed attachment stacks of0/1/2 inches, remaining heights are 15.004/14.004/13.004 inches **before movement clearance**. The actual three terms are unknown. Wheels elsewhere can rise above that plane if their spatial envelopes clear the frame; a single global height does not establish fit. Require supplier CAD at loaded/unloaded, bump/rebound, roll, yaw and worn-wheel conditions. Do not notch a girder or lower a bogie support simply to make the drawing fit.

3. Member properties and what the calculations mean

Atlas publishes the following strong-axis properties and nominal weights C3S01. Nominal3/16- and⅛-inch walls use0.174/0.116-inch design walls; nominal¼ uses0.233. Exact stock is unverified.

Candidate HSS, inches	I, in⁴	S, in³	lb/ft	Role
12×4×¼	119	19.8	25.82	Main and bolster benchmark
10×4×¼	74.7	14.9	22.42	Main-girder alternative
4×2×3/16	3.66	1.83	6.87	Cross/perimeter benchmark
4×2×⅛	2.65	1.33	4.75	Lighter comparison
3×2×3/16	1.77	1.18	5.59	Shallower comparison
6×2×3/16	10.5	3.50	9.42	Deeper comparison
3×1×⅛	0.817	0.545	3.05	Seat-attachment rail study

A500 Grade C50ksi is a candidate material basis, not an allowable-stress criterion. Obtain material traceability and engineer-selected toughness. A1085’s tighter tolerances/CVN provisions may merit evaluation, but it is not substituted into the A500 property table C3S04.

structure.py uses E=200GPa and Euler-Bernoulli elastic beam equations. Transverse supports are idealized at y±34, ends at±54, assuming developed continuity across girder nodes. It verifies force equilibrium, support displacement and the independent uniform-overhang solution. This is **not** a complete frame model: actual joint flexibility, shear distortion, deck distribution, wall/roof/equipment dead loads, fatigue, dynamic amplification, lateral loads, twist and connection limit states are absent.

Alternative local load case	HSS4×2×3/16 stress / deflection	HSS4×2×⅛ stress / deflection
4kPa uniform floor trial;16-inch tributary	13.22MPa /0.362mm	18.19MPa /0.499mm
Entire four-person row plus two15kg bench allowances onto one node	12.67MPa /0.149mm	17.43MPa /0.206mm
Two145kg adults at one outer platform edge	48.18MPa /1.795mm	66.29MPa /2.479mm

These results support investigating lighter crossmembers; they do not approve either size. The outer-edge case produces **3.680kN** on the near girder and **−0.836kN** on the opposite one. That local negative reaction identifies a required restraining/moment-transfer path in the assumed model; it is not a prediction that the whole coach overturns. Stairs, guards, concentrated equipment and boarding impacts require additional

cases.

4. Seats and floor panels need different load paths

Retain 32 places: two-person benches each side, eight rows at 32-inch pitch. Bench centres x41,73,...265 match the retained geometry. For load development only, each occupied bench is290kg passengers plus15kg assumed bench mass: **2.991kN**. Four assumed feet at row-centre±6 inches and y20/44 on each side carry0.748kN each before dynamics. No seat product has been selected.

The script transfers each foot through its longitudinal rail to adjacent transverse node centrelines. It preserves all 16 bench loads; the largest node receives3.739kN. The model's actual-seat case is separate from the conservative whole-row-on-one-node case. A single foot centered on a 16-inch HSS3×1×⅛ span produces8.51MPa and0.015mm; local bearing, fastening and fatigue remain untested.

At 0.8m/s², one 305 kg occupied bench produces 244 N ordinary longitudinal inertia. With an assumed 0.65 m resultant height and12-inch fore/aft leg-pair spacing, the vertical pair-load transfer is 520 N. This is **not a complete seat anchorage requirement**: the responsible engineer must select crash/impact, abuse, vertical and lateral criteria. Use engineered hard-point inserts/sleeves so anchors do not crush the finish, bedding or honeycomb. Wheelchair restraint and lift anchors require their own supplier loads and structural backing.

Distributing33 people at 145 kg across the 306×104-inch cabin gives **2.2855kPa**. The additional4kPa case is an analyst local sensitivity, not an invented code requirement or extra train payload. Neither uniform occupancy case is added to the explicit seated loads. The transverse beam calculation conservatively loads108 inches including edge strips; the floor-bay quantities below use104 inches clear.

Floor study span	Full-width bay load at4kPa	Required EI,1-inch strip, L/240 target	L/360 target
16in	4.294kN	7,426lbf·in²	11,139lbf·in²
24in	6.441kN	25,063lbf·in²	37,594lbf·in²
32in	8.588kN	59,407lbf·in²	89,111lbf·in²

These bending-only stiffness demands follow EI=5q b L⁴/(384δ); L/240 andL/360 are comparison targets, not accepted deflection limits. Honeycomb core shear, local indentation and supported edges need separate treatment. Plascore's published rigidity row has inconsistent unit labeling, so the calculator **does not use it as an approved EI or span** C3S02.

A145kg person on one assumed50mm-square contact produces0.569MPa;35% of a295kg occupied mobility device on one assumed20mm-square caster produces2.531MPa. Actual footprints and wheel load sharing are unknown. These are demands for the floor supplier, not comparisons to a design allowable. Obtain rolling/point-load tests and complete skin/core/edge/insert details before choosing support spacing.

5. Mass is now measurable, but not closed

Disjoint member group	Count	Modeled mass
Continuous main girders	2	780.78kg
Bolster segments, replacing ordinary nodes	6	191.29kg
Ordinary crossmember segments	72	610.77kg
Perimeter side rails	2	207.75kg
Seat attachment rail segments	84	122.67kg
Bare modeled total	**166**	**1,913.26kg**

The 166 geometric envelopes have no positive-volume overlaps. Counts are **not a cut list**: joint gaps, mitres, coping, end closures, cutting waste and fabrication tolerances are absent. Draft structure, bolster/sidebearer attachments, reinforcement, mounts, weld metal, coating, stairs/guards/lift structure and lifting/jacking provisions are excluded. Floor/body panels and bogies are outside this subtotal and must not be counted again as steel.

Keeping 16-inch spacing but changing cross/perimeter members to4×2×⅛ gives 1,660.67 kg. Also changing the main pair to10 inches gives 1,557.86 kg, before held items. A24-inch grid with the thin crossmembers and10-inch mains gives 1,424.47 kg, leaving only 75.53 kg of the original allocation before those omissions. This does not demonstrate a1,500kg finished frame. The32-inch alternatives save relatively little once mandatory bolsters and other nodes remain, while floor stiffness demands increase sharply. Do not exchange a coordinated16-inch grid for a wider one without revising panel seams, seat supports and load analysis.

6. Bolster roll support is a governing decision

Retain Phase 3B’s severe one-sided front-crowd study:81.343kN front reaction and32.578kNm roll moment. If an idealized pair of compression supports carried **all** that vertical load, their force difference would be $2T/s$, where s is their separation. At 30-inch separation, one required reaction is−2.082kN; at 36 inches, the two are76.300/5.043kN. Any vertical load carried by a central pivot reduces the load available to that pair. These are sensitivity examples, not selected sidebearer dimensions.

The bogie supplier must define centre/sidebearer/suspension load sharing, retention, roll stiffness and wheel unloading. A central pivot alone cannot be assumed to transmit this roll demand. Local HSS connection and combined-force checks must accompany member checks; a large tube and an unspecified perimeter weld do not prove the connection C3S05.

7. Required release evidence

Continue independent cabin mockups and supplier technical comparisons. Before structural cutting, obtain the real bogie interface, agreed load cases and complete floor test/allowable data. The structural engineer then resolves the 3D load path, local walls, combined bending/torsion/axial loads, fatigue, draft/braking, jacking/lifting and deflection limits; selects materials and qualified joint details; and closes the mass/CG budget. Accessible lift/securement attachments require a distinct frame review.

For a future first-chassis test, reserve instrument locations at both girder lines near bolsters, midspan and platform tips. Establish calibrated load, strain, displacement and wheel/corner-weight measurements, baseline surveys and post-test inspection. Test loads, increments, hold times and acceptance limits remain engineer-return fields. Farm labor can prepare mockups and approved work areas; it should not fabricate unreviewed structural joints or improvise proof loads. This issue provides engineering input and review geometry, not permission to carry passengers or procure the listed tubes.

04 | Floor, body, boarding and service interfaces

PF-TRAIN-S3C-R0 · 17 October 2026 · Preliminary assembly studies; not fabrication approval. Retain the 306 × 104 in clear cabin, 312 × 110 in outside body and provisional finished floor 28 in above rail. Track targets 0% longitudinal grade. The 32-place regular layout and accessible 11 fixed seats plus two wheelchair positions remain mockup choices; seat anchors must follow the accepted trial. This issue develops physical assembly interfaces without releasing unsupported joint sizes.

Floor stack and support plane

The previous panel-only frame elevation omitted floor finish and installation layers. Use a common finished-floor datum; changing materials must move the supporting structure or trigger a coordinated floor/door/stair change.

Floor option	Product thickness	Trial total thickness	Steel bearing elevation	Panel + finish mass
3/4 in Plascore + Sirius NT	21.30 mm	25.30 mm	27.0039 in	146.79 kg
3/4 in Plascore + Sirius MK	25.25 mm	29.25 mm	26.8484 in	176.56 kg
3/4 in Coosa + Sirius NT	21.30 mm	25.30 mm	27.0039 in	220.84 kg
1 in Plascore + Sirius NT	27.65 mm	31.65 mm	26.7539 in	154.36 kg

The extra 4 mm is an analyst packaging reservation: 1 mm finish adhesive plus 3 mm support isolation/bedding. Neither is an installation specification. Supplier-qualified materials, compression, tolerances and bond geometry replace these values before release. The calculation file also includes MK combinations for both alternate panels. NT is 2.25 ± 0.15 mm and 2.20 ± 0.15 kg/m²; MK is 6.20 ± 0.30 mm and 3.65 ± 0.30 kg/m². Their stated thickness is already the finished product, not a backing to add twice. [C3B01-03]

The floor panel covers 238.33 ft²; interior finish covers 20.5316 m². These masses exclude adhesives, isolation, edge closures, inserts and supporting steel. The 1 in honeycomb alternative adds only 7.57 kg of panel, but its thickness cannot establish adequate span or concentrated-load performance. Coosa adds 74.05 kg relative to the 3/4 in honeycomb case; equivalent thickness does not mean equivalent stiffness or screw capacity.

Panel seams now coordinate with the frame

Use transverse seams at x = −3, 45, 93, 141, 189, 237, 285 and 309 in. These match the proposed frame support nodes. Two purchasing studies are possible:

- **Whole-width panels:** six 48 × 110 in pieces and one 24 × 110 in piece. Plascore explicitly offers AA3.6-80 up to 5 × 12 ft; rotate these pieces within 60 × 144 in blanks. Seven blanks give 420 ft² purchased versus 238.33 ft² installed, 56.75% geometric yield before trimming. Requesting custom-cut parts could reduce waste; no quotation exists.

- **4 × 8 ft stock:** divide each transverse row into 21/68/21 in strips, with longitudinal seams at y = −34 and +34 in, directly over the proposed main girder centers. There are 21 pieces. A demonstrated, nonoptimized nine-sheet nest yields 82.75% before kerf. The source file gives every piece and blank position.

Both are nominal coverage studies: real joints reduce piece dimensions and need edge closures, bearing width and fastener access. The old eight-sheet quantity example placed seams at y = −7 and +41 in without matching supports; it must not become the production cut list. The 16 in maximum frame grid is also a study, not a certified honeycomb span. [C3B01]

Seat legs and wheelchair anchors need direct load paths into released steel rails/backing. Use supplier-designed hard points and compression sleeves where needed; thin skins, foam, finish and compressible isolation cannot be assumed to carry bolt clamp loads. Plascore offers potted, through-fastened and framed-edge constructions, but publishes no project insert rating. Fastener diameter, grade, pitch, torque and replacement method remain held. Make seams and critical anchors inspectable before concealing them. [C3B05]

Dimensioned wall and roof zones

At the right wall, the clear interior face is y = 52 in. A 1/4 in liner occupies 52…52.25, a 2.25 in construction cavity occupies 52.25…54.5, and a 1/2 in exterior panel occupies 54.5…55. Mirror this at the opposite wall. The cavity includes posts, insulation, thermal breaks, mounting/seal interfaces and services; it is not entirely usable insulation space. A 13 mm insulation trial leaves 44.15 mm before structure and services are deducted. The chosen insulation’s thin-product fire claims cannot be extended to a thicker multilayer assembly. [C3B07]

The finished ceiling remains 2,200 mm / 86.6142 in above the cabin floor, at rail z = 114.6142 in. A flat section with 1/4 in liner, 2.25 in cavity and 1/2 in outer panel reaches z = 117.6142 in. Roof crown, weather cap, curb and equipment add height. This is a coordinated section reservation, not the final curved roof or overall loading gauge. A 415 mm-high Konvekta 7403 above this flat proxy reaches 133.9528 in before its mounting allowance; its 520 kg needs engineered roof-to-frame support. A 15 in unit below the ceiling would leave only 71.6142 in headroom, so that placement remains rejected. [C3B13]

The nominal 78 in door opening leaves an 8.6142 in header zone below the ceiling. Door operators, lighting and ducts must stay outside the actual clear opening and aisle envelope. Roof access panels require a demonstrated removal path, lifting arrangement and fall-protected maintenance access. Do not create a service opening through a structural bow without redesign.

Assembly details to qualify

Develop one representative floor/wall corner with a panel seam, seat hard point, door threshold and accessible inspection cover. Add a separate window-to-roof sample incorporating the actual frame, glazing retainer, seals and drainage path. The samples should use production substrates and coatings; their final dimensions follow the selected hardware.

Engineer body attachments for braking, lateral acceleration, racking and applicable wind/roof loads; do not credit decorative panels as structure. Separate the functions: structural support carries loads; retained panel joints handle differential movement; exterior flashing and seals exclude water; removable trim conceals joints while allowing inspection. Provide sealed honeycomb edges, weather-side overlaps and drain paths outboard of the body/frame cavity. Electrical bonding must be intentional across aluminum/steel isolation, not left to incidental fastener contact. Locate condensate outlets away from stairs, bridges, brakes and electrical enclosures; verify flow with the vehicle at its permitted roll/pitch attitudes and drains partly obstructed. Level track does not create drain slope.

Sikaflex-252 is a candidate vehicle-body adhesive, not an automatic floor-finish adhesive. Its 10-35°C application range and requirement for tests on actual substrates inform a controlled shop process. Record surface preparation, coating lot, batch, expiry, temperature, humidity, assembly time, cure and coupon failures. Its typical 2.5 MPa lap-shear value is not an allowable joint stress. Qualify peel, fatigue, thermal movement, wet exposure and repair; do not conceal an unverified structural bond. [C3B06]

Gerflor’s installation guide calls for clean, dry, supported substrates, 15-30°C installation/storage and 24 hours flat relaxation. Its generic water-based acrylic option is for absorbent substrates; do not assume it

suits coated aluminum. Use an accepted adhesive, seam process and cleaning product. Trial muddy footwear, wet cleaning, abrasion and threshold transitions on the actual assembly. Obtain complete material/assembly fire requirements from the responsible professionals; individual FMVSS, ASTM or EN claims do not establish vehicle approval. [C3B03-04]

Platforms and stairs: usable envelopes, unresolved hardware

The front platform remains x = −47…−3, y = ±55 in. Reserve a 36 in clear stair width along x = −43…−7, deploying outward from y = 55. Only 8 in of platform length remains for end framing and posts, so this envelope still needs real guard/handrail fit.

Let the actual landing elevation above rail be zG and the number of risers be N. Uniform study rise is (28−zG)/N; a trial 11 in going gives outward run (N−1)×11. At zG = 0 and N = 4, the example has four 7 in rises and three treads projecting 33 in outboard, to y = 88. With the same count and landing zG = 4, rise becomes 6 in. **These are geometry examples, not adopted vehicle-code dimensions.** The calculation matrix compares three landing heights and three riser counts.

The side stair keeps boarding away from the intercar coupling. Develop continuous hand support, contrasting nonslip edges, lighting, self-closing controlled gates and positive travel retention. Check the complete deployed and stowed sweeps, suspension motion, track curves and debris clearance. Do not assume a folding design fits within 110 in while stowed. Final step, guard, hinge, latch, weld and bolt details await the vehicle boarding criteria and engineered prototype. No intercar passenger walkway is implied.

Accessible boarding and rescue

Continue with a positively retained bridge at both stations, using the 42 in side opening and preserving the 36 in route. A 36 in clear bridge reservation is x = 12…48. For study only, choose a least-practical-slope target no steeper than 1:12: a 36 in working horizontal run accommodates 3 in vertical mismatch geometrically. Working run is not the structural span or bearing overlap; those remain unknown. Survey, empty/loaded suspension, roll, alignment tolerance and train immobilization determine the actual interface. Access Board provisions are research references until applicability is resolved. [C3B08-09]

The optional Braun lift still requires at least a 45.85 in clear doorway; a 48 in trial opening alone did not fix its inward reservation. Moving the 60 in trial turning circle from (30,6) to (30,3) yields a nominal 1 in separation from the conservative lift boundary y = 34. This removes the mathematical overlap only. Exact stowed CAD, wall/door hardware, tolerance, the connection to the offset route and real-device swept paths must establish usability. Keep the bridge arrangement as the primary space study. [C3B10]

Rescue must work with dead vehicle power and a blocked lift side. A mobile lift still needs reachable firm support, compatible bridging and a demonstrated route beside every permitted stopping location. The rail-specific Mobilift TX is rated 600 lb: it fails even the 295 kg / 650.36 lb screen of a 145 kg person plus a 150 kg device. A 750 lb Genesis STAAGE passes that arithmetic but fails 440 kg with an attendant or 445 kg when one device uses the entire 300 kg train device allowance. A 1,000 lb rating also fails the 590 kg person/heavy-device/attendant case. Device allocation is not an operating weight limit. Choose the admitted-device and assistance basis explicitly, then select and demonstrate a complete rescue system; no currently researched portable unit is accepted for all cases. [C3B11-12]

Farm crew can build templates and ground mockups, record dimensions/masses and perform trained, released finishing processes. Structural anchors, critical bonding, stairs/guards, boarding equipment, controls and acceptance tests require the appropriate specialists. These studies authorize further engineering and sample development, not passenger operation.

05 | Brake controls and intercar connection interfaces

Development decision

Retain hydraulic service braking and develop a locally powered, two-zone coach package: one zone for each two-axle bogie. The preferred study uses spring-applied, hydraulically released actuators, with modulated release pressure for service braking **only if the brake supplier accepts the complete repeated-service duty**. Otherwise retain hydraulic service through active calipers and provide a separately justified emergency/parking arrangement. A complete automatic-air package remains an alternative if a locomotive/brake integrator can supply it with better support.

Spring application gives the car a local source of braking energy. It does not solve every failure: a blocked release line, trapped auxiliary pressure, seized actuator or incorrect valve arrangement can prevent application. Two hydraulic release circuits on a caliper do not constitute two independent stopping paths.

The common study remains 14,500 kg locomotive, three 13,200 kg regular coaches and one 11,000 kg accessible coach: **65,100 kg**, on **0% longitudinal grade**. These are proposed engineering load cases, not rated masses. The first coach must demonstrate unloaded as well as ballasted behavior before any train-level acceptance.

Documented hardware and a 2028 purchasing issue

Knorr-Bremse's HSC1P10N remains the compact caliper candidate: the manufacturer lists 20 kN brake force per disc, two release circuits, a maximum 45 mm disc width and approximately 15 kg. Its public sheet does not resolve the force convention, operating pressure, mount geometry or full trailer-axle service duty. Do not compare that 20 kN directly with wheel/rail force. [C3C01]

Hanning & Kahl's EHM reference integrates independent activation of two bogies, with 24 VDC power, a 48 VDC train-control wire, proportional regulation per bogie and a **160 bar permissible system pressure**. Its published envelope is 1100 × 510 × 500 mm and mass approximately 90 kg. The 160 bar value is not this project's caliper release setting. [C3C02]

The manufacturer now advertises ECU controllers replacing HEYxPxG and HEY-C from **2028**, matching the intended first-train year. The ECU page lists nominal 24-110 VDC and hardwired, CAN, MVB and Ethernet interfaces. Obtain the actual delivery configuration, current demand, service support and controlled safety documentation before choosing a legacy controller. [C3C03] The new HCU is configuration dependent; its advertised adjustable relief valves and filtration do not establish a complete emergency circuit or selected pressure. [C3C04]

The legacy controller page's prose and certification table differ in how they describe SIL capability. This package assigns no SIL or safety certification to the train; certificate scope, configuration, application conditions and integrator review are unresolved. [C3C06]

Functional hydraulic and electrical architecture

Drawing H01 should show **functional blocks**, not valve symbols implying completed plumbing. D01 driver service demand enters BC01 coach control. BC01 requests front/rear working release pressure from H01/H02. Each zone serves two candidate spring actuators, SA01-SA04. A higher requested spring-brake force generally requires lower working release pressure; use the supplier's actual nonlinear force curve, including wear, temperature, hysteresis and residual drag.

EM01 is a separately monitored emergency/breakaway permit function. It commands EV01/EV02 to remove working release energy and removes locomotive traction permission through TR01. EV01/EV02 describe required application functions; **the number, arrangement and deenergized position of valves remain held**. A simple single dump valve is not declared sufficient. The supplier must demonstrate application

despite the agreed credible trapped-pressure and valve-failure cases.

PW01 supplies protected local brake power. Nominal 24 V is only an integration study; it cannot be substituted for an EHM's documented 48 V control wire without an approved interface. Hotel AC and ordinary passenger controls must be segregated from brake power and emergency commands. Required returns include inrush, continuous and holding current, undervoltage behavior, protection coordination, EMC, bonding, fault containment and any reserve energy. Neither frame nor running rails are assumed to be normal electrical return conductors.

The preferred scheme has **no intercar hydraulic service hose**. The intercar functions are service demand/data, independently monitored emergency/breakaway continuity, protected control-power distribution where required, and protective bonding. Connector pinouts, wire gauges, channel count and termination arrangement are unselected. Loss of one ordinary data network must not be the only means of detecting a disconnected coach.

AR01 is a keyed maintenance/recovery release function using the actuator's auxiliary circuit. Hanning & Kahl documents this second circuit specifically as an independent means of releasing spring brakes. [C3C05] During normal operation it must be inactive and its state monitored. Manual/mechanical release and hydraulic isolators require a controlled recovery procedure, movement restraint and restoration proof; a reset command must not silently release a parked car.

Commands and fault responses

Condition	Required design response and verification
Normal service	Valid demand and load information produce proportioned braking; compare actual pressure, actuator movement and speed response with demand.
Emergency or local control-power loss	Remove traction permission; apply and retain the approved brake force locally; demonstrate mechanical response and deliberate reset.
Service data lost or stale	Reject invalid commands and invoke the defined stop through a separate emergency function; timeout is an approved supplier-return value.
Emergency wire open	Demand emergency in each affected portion. A lamp alone does not prove detached-car braking.
Emergency wire short/cross-short	Detect faults capable of masking an open circuit. Merely installing two wires or channels is not proof of independence.
Working hydraulic hose rupture	The spring zone tends toward application as pressure falls, potentially creating asymmetric braking. Measure the transient; detection must initiate the defined train response.
Line blocked or release valve stuck	Release pressure may remain trapped. Require a demonstrated alternate application path or reject the architecture.
Coupler breaks while cables remain connected	Electrical continuity may persist. Engineer a separate separation-detection function and controlled jumper breakaway sequence before cable tension causes damage.
Parking/recovery	Hold without locomotive energy; prove the parking load and duration. Auxiliary release or isolation must inhibit normal traction permission.

The JSON failure matrix adds caliper/pad defects, bogie loss, contamination, wheel-slide/speed disagreement and controller reboot. Wheel-slide protection at these low speeds, and transition into parking hold, require OEM validation. A stationary axle alone does not prove that the train has stopped.

Quantified response and brake interfaces

The retained analyst deceleration targets, 0.50 m/s² service and 0.80 m/s² emergency, imply 32.55 and 52.08 kN total wheel/rail force. Regular-coach emergency demand is 10.56 kN, or 2.64 kN per axle under the provisional four-axle allocation; accessible demand is 8.80/2.20 kN. Actual axle loads and brake balance supersede equal allocation. Required axle torque is force multiplied by actual loaded wheel radius; neither radius nor disc geometry is selected.

At 7.5 mph the train's translational energy is 365.90 kJ, before rotating components. Maintaining the previous **3 s equivalent delay** produces an ideal 17.084 m emergency stop. One additional unbraked second adds **3.353 m** at that speed. Delay must therefore be measured from the originating command to actual force, including fault recognition, transmission, valve action and buildup.

New reproducible calculations model 28 service/emergency cases with linear force ramps and 112 connection-force records. These compare assumptions, not tested performance:

Hypothetical 7.5 mph emergency response	Calculated level distance
All force begins after 3 s, instantaneous buildup	17.084 m
All vehicles begin after 0.5 s; 1.5 s buildup	11.142 m
Vehicle delays 0.3/0.6/0.9/1.2/1.5 s; 1.5 s buildup	12.304 m
Same staggered case with 5% effective rotating inertia	12.662 m

The shorter distances do not replace the retained reference or establish operating limits. Neither delay profile is measured. The rotating-inertia sensitivity holds brake torque divided by wheel radius fixed; some torque decelerates rotating components, so actual wheel/rail force is lower than that equivalent force. The model excludes slack, impact, adhesion variation, fade and wheel-slide regulation. Closed-form and timestep checks verify arithmetic only.

A fixed 10.56 kN regular-coach brake force would produce approximately **1.50 m/s²** at the 7040 kg tare allocation instead of 0.80 m/s² at the upper case, requiring ideal adhesion about 0.153. This is a hypothetical force setting, not an actual spring selection. It makes unloaded braking, load correction and passenger jerk essential selection issues.

Connections and frame attachments

Carry a railway semi-permanent connection with controlled retention, articulation and energy absorption. Dellner documents a damper, pivot anchor and cable support; configured ratings and dimensions remain absent. [C3C07] Voith advertises up to 300 kN tension/compression for the 430/530 family, which does not rate our selected configuration or coach attachment. [C3C08]

Retain 58 in pivot span, 30 in platform gap and 14 in setbacks solely as geometric studies. Do not shorten OEM drawgear or replace it with a trailer tongue. The load path is connection assembly → engineered anchor/diaphragm → both longitudinal frame paths → vehicle structure. Resolve longitudinal force, lateral/vertical components, eccentric bending and torsion through the actual articulation envelope; platforms and guards are not drawgear anchors.

Phase 3B launch force at the first joint reaches 12.55 kN tension at assumed 0.15 m/s² acceleration and 0.010 starting resistance. Its locomotive-only braking study, assuming adhesion 0.15 and all 14,500 kg of locomotive mass braked, gives 16.58 kN compression there. New staggered braking produces rigid-consist compression peaks of 4.39/6.28/6.05/3.71 kN at successive joints. Reverse travel changes signs for the same force allocation. None includes slack impact or establishes proof/fatigue ratings. Actual drawgear stiffness, stroke, damping, collision/coupling cases and attachment offsets are needed before sizing pins, plates, bolts or welds.

First-coach evidence and release gates

The saved interface register defines 11 boundaries and 10 supplier return forms, with blank evidence, accepted limits and reviewer fields. Test points include front/rear working and auxiliary pressure, four actuator travels, four axle speeds, independent ground speed, temperatures, power, demand timestamps, connection force and stroke.

Proceed through controlled schematic review, guarded bench testing, restrained unoccupied coach testing, coupled static checks, then contained ballasted running trials. The responsible engineer sets numeric pressure, torque, response-time, stopping, parking, leakage, thermal and connection acceptance limits **before each test**. Use synchronized calibrated instruments and recorded uncertainty. Initial breakaway tests simulate disconnects on a secured car; intentional moving separation needs a separately engineered containment plan.

Farm crew can assist with identification, accessible cable supports and inspection records under released drawings. Brake plumbing, controls programming, hydraulic cleanliness, safety analysis, connection fabrication and acceptance require the appropriate specialist. No supplier has been contacted and no component is released for ordering by this package.

06 | Fabrication development and first-article records

PF-TRAIN-S3C-R0 · 17 October 2026 · Preliminary production planning; not released for fabrication. The immediate physical task remains the ground-level interior mockup. No evidence of completed mockup trials, approved bogie interfaces or signed fabrication drawings has been received. This package makes the first coach's work sequence, records and release conditions concrete without treating those missing decisions as completed.

Design and purchasing gate

Develop one regular coach as the first article. Its two complete bogies form the first pair of the eight-bogie train lot; do not count them as additional production units. The accessible vehicle needs separate boarding/evacuation and structural verification. Its success cannot be inferred from the regular first article.

The current frame model contains **166 pieces**: two main girders, six bolster segments forming two transverse assemblies, 72 crossmember segments, two side rails and 84 seat-attachment segments. Its **1,913.26 kg bare modeled steel** already exceeds the former 1,500 kg frame allocation by 413.26 kg, before joints, draft details, boarding attachments, weld metal and coating. This is a modeling quantity, not a cut list. Resolve the weight/structure decision before ordering steel. Reconcile actual tare and CG after each major assembly; do not consume the conservative upper load case as a new weight allowance.

Required controlled documents are C3-G01/F01/D01/A01/H01/C01/J01/V01 plus the released BOM/cut schedule, weld map/WPS index, inspection plan, coating/bonding specification and lifting/support plan. The included C3 sheets remain preliminary. Their titles do not imply a released revision exists. Procurement gates require configured OEM drawings, compatible loads and interfaces, traceable materials and agreed acceptance records - not gauge matching alone.

Welding basis to close with the fabricator

Evaluate **AWS D15.1/D15.1M:2025** first because its published scope concerns North American railcars, locomotives and components. AWS also publishes an errata link. The scope includes thinner-than-1/8-inch base metals; do not automatically place every thin body component under an unrelated shop practice. [C3F01]

AWS **D1.1/D1.1M:2025-AMD1** covers structural steel welding, while **D1.2/D1.2M:2026** is the current structural aluminum edition. The responsible welding/vehicle engineer must state which applies to each assembly and reconcile overlapping provisions, fatigue requirements and inspection criteria. We reviewed public publisher scope pages, not the complete purchased standards. No standard here is asserted to be automatically required by Nevada law, and compliance with a welding standard alone does not approve the vehicle. [C3F02-03]

For a supplier proposing EN 15085, check its current certificate scope, location and processes through the recognized rail welding register and obtain the project-specific quality plan. Do not declare that certification interchangeable with the selected AWS contract requirements. [C3F04]

The author must return joint geometry, weld size/length, load category, WPS qualification basis, welder qualifications, consumable, root preparation, permitted tacks, preheat/interpass/heat-input controls, inspection timing/extent and repair rules. Those fields are blank deliberately. An inspector checks against the released criteria; the inspection signature is not a substitute for structural design responsibility.

Datum and jig plan - sheet C3-J01

Use shop references **A** = frame floor-support plane zF=0, **B** = longitudinal center plane y=0, **C** = front bogie crossline x=45 in. These are conceptual coordinate references, not a completed GD&T specification. Keep the project origin x=0 at the cabin's front inside wall. A surveyed tooling target may sit away from A; record its transfer offset. A tooling surface is not automatically the rail datum.

Study check	Nominal geometry
Frame envelope	x=−47…353; y=−55…55 in
Main girder centerlines	y=−34/+34 in
Bogie center stations	x=45/261; spacing 216 in
Envelope diagonal	414.849 in
Girder-end control diagonal	405.739 in
Bolster/girder intersection diagonal	226.451 in
Cabin diagonal, 306×104 in	323.190 in

Diagonals are calculated nominals, **not tolerances**; equal diagonals alone do not demonstrate flatness, twist or interface alignment. Record both girder lines at x=−47, −3, 45, 105, 153, 201, 261, 309 and 353: **18 measurement points**. Measure longitudinal/transverse position and height using a competent shop's surveyed instrument method. Record instrument/calibration references, uncertainty, temperature, support condition and inaccessible-point offsets. NIST distinguishes measurement traceability and uncertainty from simply having a calibration label. [C3F06]

Survey at jig setup T0, dry fit T1, tack T2, named weld groups T3, cooled/unclamped T4, and before/after ballast T5. The editable record contains 126 initial point/stage rows; duplicate T3 for each approved weld group. The engineer supplies production tolerances and the measurement method must resolve them. The mockup's ±1/8-inch marking target is not transferred to the frame.

The 25.30 mm body floor-stack study would put A at rail z=27.003937 in if the finished floor stays 28 in. This remains a packaging trial. Jig support locations, clamp forces, fixture foundations, turning gear and lifting attachments require a rated temporary-work design; the drawn measurement stations do not authorize improvised stands.

Assembly and inspection sequence

Each row has dependencies, input drawing IDs, output record IDs, responsible roles and blank actual results/signatures in the editable traveler. Steps 04-23 start only after the fabrication release and accepted parts are available.

Work package	Work and output
01	Tape full-size interior; trial six adults; record MOCKUP-01.
02-03	Close engineering; release documents, technical purchases and receipt plan.
04-06	Survey rated jig; match certificates; mark heat/lot to pieces; cut and inspect released preparations.
07-09	Fit bolster/draft subassemblies; inspect tacks; weld to mapped procedures; inspect before obscuring faces.

Work package	Work and output
10-12	Dry-fit main frame, tack and survey; execute approved weld groups; cool, release restraints, survey and complete NDT.
13-15	Trial-fit actual hard points; qualify body-joint/bond specimens; inspect hidden work, then prepare/coating operations.
16-18	Install bogies/brakes/connections; dry-fit body; assemble panels, insulation and liners with closure records.
19-20	Fit doors, glazing, benches and boarding; complete electrical/HVAC, drainage and controlled leak checks.
21-23	Weigh and test secured ballast; controlled rolling/braking; inspect, reconcile as-built records and decide production release.

Transfer material identity before removing original markings; isolate nonconforming stock. Use approved marking locations so identification does not create a fatigue defect. Record measured thickness separately from nominal stock and structural design thickness. Do not substitute a similar-looking section or drill/notch a main girder for late hardware.

Develop a numbered weld-group map that balances corresponding left/right and front/rear joints while preserving required access. TWI identifies tack planning, qualified incorporated tacks and balanced sequences as distortion-control tools; it does not supply this frame's settings. Recheck geometry after the specified groups and after cooling/free release. Do not force the finished frame into tolerance with its jig, or heat-straighten it without an accepted repair procedure. [C3F05]

Keep every deviation visible through disposition, repair, reinspection and as-built update. NDT method, coverage and acceptance depend on the joint/material and responsible engineer; no arbitrary percentage or universal test method is assigned. Inspect primary joints and attachments before coatings or liners hide them.

Body assembly, ownership and records

Build the representative floor/wall/roof/window joint with the actual coated steel, isolation layer, aluminum and panel materials. Coupon plans must establish surface preparation, bond geometry, conditioning, cure, failure mode and acceptance. Sika requires actual-substrate/application verification; published adhesive strengths are not vehicle design allowables. Its general guide also highlights coating/substrate effects. [C3F07-08]

Separate intentional electrical bonding from galvanic isolation. Keep defined bonding/contact surfaces, precision fits and adhesive zones protected during painting. Record coating batches, conditions, film thickness and cure. Dry-fit drainage, sealed panel edges, inserts, wires and service access before permanent closure. Do not load bonded assemblies until their specified cure release. Photograph each hidden layer and weigh representative modules.

Farm crew may undertake the ground mockup, receiving assistance, labels/photos, nonstructural templates and trained finish/panel installation after process approval. Qualified specialists own structural cutting/welding, NDT, wheel/bearing work, lifting/turning, brake integration and electrical/HVAC installation. Responsible engineers own tolerances, structural/interface approval and test acceptance. Training can expand a crew member's assigned scope; ownership must be documented rather than assumed from job title.

The forms include material, weld, bond, fastener, coating and deviation records; **18 interface-closure rows**, six procurement holds and 13 inspection stages. A blank limit, document revision or required release signature means **HOLD**, never an automatic pass. First-article acceptance remains separate from authorization to carry the public.

Relative working-day plan

Allow **2-3 working days now** for the interior mockup: two crew members spend 4-6 hours marking/checking plus 4-6 hours preparing/moving templates; six adults participate 1.5-2 hours; one recorder spends 2-3 hours. This is **27-39 person-hours**, a transparent planning allowance.

After design release, all parts accepted/on hand and specialist availability confirmed, the conservative sequential work-package allowances total **40-79 working days**, one principal two-person fabrication crew plus specialist visits. This reserves time against the modeled 166 pieces and assemblies; it is not measured production productivity or a fabricator quote. Engineering/OEM responses, supplier lead times, procedure qualification, required cure/conditioning, failed tests/rework and authority decisions remain additional unbounded gates. No 2028 delivery promise follows from this schedule.

Five 18-point surveys at four minutes/point with two people, five half-hour two-person setups and four hours of report review total **21 person-hours**. Additional weld-group and ballast surveys add work. Production labor remains unpriced/unquantified until the shop returns actual joint lengths, procedures, fit-up hours, deposition rates, inspection staffing and curing/queue times.

Sources were checked 17 October 2026. Full URLs and the distinction between publisher facts, project assumptions and unresolved author returns are recorded in fabrication.json; this workflow does not establish a released weld procedure or construction approval.

07 | Coordinated model and drawing use

The eight C3 sheets and the 3D assembly studies use the same research inputs. The 3D view helps inspect the floor support pattern and part relationships; it does not demonstrate strength, suspension clearance, stability or service access. Hollow-section envelopes omit manufacturing corner radii and final end preparations where those details are not selected.

Sheet	Purpose
C3-G01	Controlled first-coach geometry and interface map.
C3-F01	Frame grid, member/bolster sections and support-height study.
C3-D01	Floor, wall and roof construction interfaces and panel seam support.
C3-A01	End platform, stair and accessible bridge interface studies.
C3-H01	Functional hydraulic/control architecture and test points; not released plumbing.
C3-C01	Connection load path, articulation and frame attachment returns.
C3-J01	Jig datums, measurement scheme and assembly sequence.
C3-V01	First-article instrumentation, records and release gates.

The source package contains the input coordinates, model generator, calculation scripts, editable drawings and workbook authoring source. PDF and SVG are review sheets; paper-space DXF and full-size model DXF have different units/scales as explained in the drawing README. STEP solids use millimetres; browser mesh coordinates use inches. STEP does not include a native parametric feature tree.

08 | Practical next work and release boundaries

When	Work that can proceed	Required evidence before the next action
Now	Ground-level occupied layout trials; stable mockup fixtures; dimensional surveys; material samples; supplier/engineer review of prepared technical returns.	Record observations and proposed changes. Obtain separate authorization before external outreach or purchasing.
Before steel purchase/cutting	Close running-gear support height and loads; choose a load/standards basis; update frame/body mass; resolve connections and wheel/rail fit.	Responsible engineer releases coordinated member, joint, tolerance and inspection documents. Manufacturer responses must identify exact configured hardware.
Before chassis assembly	Accept material certificates, jig layout, approved welding documents and qualification scope.	Released fabrication traveler identifies who inspects each hold point and which records must be retained.
Before installing body/equipment	Measure chassis, reactions and interfaces; reconcile deviations; verify access and fastening requirements.	Accepted chassis records and closed interface deviations. Preserve inspection access and corrosion/weather protection.
Before moving tests	Complete hydraulic/electrical/static checks and secure test ballast using an engineered method.	Responsible engineers approve test track, instrumentation, procedures, fault cases, exclusion zone and numeric acceptance limits.
Before repeated coaches/passengers	Close first-article defects and reconcile measured mass, loads, stops, heat, clearances and evacuation.	Controlled production release and separate operating/commissioning approval. Study calculations alone do not satisfy these gates.

This phase does not authorize steel fabrication simply because a candidate member appears in a model. It gives engineers and builders a concrete arrangement, quantified comparisons and records to complete. The first physical task remains the occupied mockup; the next design input is an engineered running-gear interface that fits the required body and loads.

09 | Source register

Primary publications and retained project calculations are identified below and in the editable register, with access dates and evidence limits. Published data apply only to the stated product/configuration. Catalogue presence does not confirm price, stock, lead time, project applicability or current ability to supply this vehicle.

C3S01 | Atlas Tube ASTM A500 rectangular HSS properties, pp2,3,5,6

Accessed 2026-10-17

Exact nominal mass, design wall, area, strong-axis I and S for seven study sections; page5 and6 screenshots visually inspected.

Limit: Published section properties only; historical table remains manufacturer-hosted. No stock, strength acceptance, weld or rail approval.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3S02 | Plascore Base Panels technical table

Accessed 2026-10-17

AA3.6-80 at0.75in weighs0.94lb/ft2;0.020in aluminum skins; typical stabilized compression2.54MPa; stiffness table46858lbf*in2 with per-unit-width heading and inconsistent SIunit label; test span20in/specimenwidth3in.

Limit: No allowable panel span or concentrated load. Published rigidity not used until manufacturer clarifies units/normalization and applicable construction.

https://plascore.com/composite-panels/base-panels/

C3S03 | Gerflor Tarabus Sirius technical data sheet

Accessed 2026-10-17

NT2.25+/-0.15mm and2.20+/-0.15kg/m2;MK6.20+/-0.30mm.

Limit: Product dimensions only; adhesive/bedding values are explicit analyst allowances, not supplier prescriptions.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

C3S04 | Steel Tube Institute HSS Member Selection Tools

Accessed 2026-10-17

A500GradeC50ksi; nominal-versus-design wall distinction; domestic production is not distributor stock. A1085 includes tighter wall tolerance and CVN requirement.

Limit: Structural-building design resource, not vehicle code selection. Exact grade/toughness/stock require engineer and supplier confirmation.

https://steeltubeinstitute.org/hollow-structural-sections/hss-member-selection-tools/

C3S05 | Steel Tube Institute: HSS-to-HSS connection with shear, axial and in-plane moment

Accessed 2026-10-17

Primary technical example identifies combined member/joint actions and local wall limit states; no weld size transferred.

Limit: Technical principle only; project connections have different geometry, cyclic loads and load paths.

https://steeltubeinstitute.org/resources/hss-to-hss-connection-with-shear-axial-and-in-plane-moment-example/

C3S06 | Retained Phase3B running-gear/frame calculations

Accessed 2026-10-17

Unchanged x45/261 supports, main girdersy±34,13.2t regular/11t accessible uppercases;81.343kN asymmetric front-bogie reaction and32.578kNm roll moment.

Limit: Prior preliminary model, not physical tests or supplier ratings.

C3B01 | Plascore - Base Panels

Accessed 2026-10-17

AA3.6-80 typical weights: 0.75 in = 0.94 lb/ft²; 1 in = 1.01; 0.5 in = 0.84; 0.25 in = 0.79. Standard 48 × 96 in sheets; AA3.6 and AA5.2 available up to 5 ft wide × 12 ft long. Open edges.

Limit: Typical data are not approved spans or assembly ratings. Exact usable blank dimensions, tolerances, quote and availability remain unconfirmed.

https://plascore.com/composite-panels/base-panels/

C3B02 | Coosa - Bluewater Panels Imperial data sheet

Accessed 2026-10-17

Bluewater 26 density is nominally 26 lb/ft³; 4 × 8 ft × 3/4 in sheet approximately 52 lb. Fiberglass-reinforced polyurethane; standard product FMVSS 302 statement.

Limit: Exact fire-rated variant differs. No train floor span, attachment or complete fire assembly accepted.

https://coosacomposites.com/wp-content/uploads/2024/10/CoosaComposites_BluewaterPanels-Imperial.pdf

C3B03 | Gerflor - Tarabus Sirius technical data

Accessed 2026-10-17

Sirius NT: 2.25 ± 0.15 mm and 2.20 ± 0.15 kg/m². MK: 6.20 ± 0.30 mm and 3.65 ± 0.30 kg/m². Published R10 slip classification, FMVSS 302 and ASTM E648 Class 1 statements.

Limit: Wet farm-dirt behavior, floor adhesive and complete vehicle fire assembly remain unapproved.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

C3B04 | Gerflor - Tarabus Installation Guidelines,December2021

Accessed 2026-10-17

Install/store at 15-30°C; relax flat for 24 hours. Clean, dry, secured substrate; metal-floor preparation, compatible adhesive and seam process. Generic water-based acrylic advice applies to absorbent substrates.

Limit: No automatic acrylic approval over coated aluminum. Confirm exact product and current manufacturer installation process.

https://cdn.gerflor.com/media/2/37754/installation%20guide%20tarabus%20-%20eng.pdf

C3B05 | Plascore - Value Add panel construction

Accessed 2026-10-17

Manufacturer offers framed panels, edge closures, blind potted/rivnut attachments and through-fastened inserts, plus assembly and quality control.

Limit: Construction approaches are not project insert ratings or released dimensions.

https://plascore.com/composite-panels/value-add/

C3B06 | Sika - Sikaflex252 US data sheet04.01,April2025

Accessed 2026-10-17

Moisture-curing vehicle adhesive; application 10-35°C; actual-substrate tests required. Typical lap shear 2.5 MPa, uncured density 1.20 kg/L; electrically nonconductive.

Limit: Typical strength is not allowable stress. Actual substrates, creep, peel, fatigue, fire and corrosion need qualification. No floor-finish adhesive approval inferred.

https://industry.sika.com/dam/dms/us01/o/sikaflex-252.pdf

C3B07 | Armacell - ArmaFlex Rail SD technical specifications

Accessed 2026-10-17

Conductivity ≤ 0.040 W/(m·K) at 0°C; E162/E662 statement limited to 3-13 mm; vapor diffusion factor ≥ 5,000; UV protection and specified adhesives required.

Limit: Density not adopted. No extension to thicker/multilayer fire assemblies; 13 mm trial is not proven Southern Nevada insulation design.

https://www.armacell.com/en-NL/product/technical-specs/armaflex-rail-sd

C3B08 | US Access Board - ADA Chapter10 amusement rides

Accessed 2026-10-17

Section 1002.4.3 addresses direct boarding gap coordination and a bridge exception referencing 1192.83(c); side entry and companion provisions also apply when scoped.

Limit: Project applicability unresolved. Separate accessible coach is not thereby approved as complete compliance.

https://www.access-board.gov/ada/chapter/ch10/

C3B09 | US Access Board - Light Rail §1192.83(c)

Accessed 2026-10-17

Section 1192.83(c) covers bridge/ramp load, surface, width, thresholds, edge barriers, slope, retention and stowage; permits station devices subject to conditions.

Limit: Reference pending jurisdiction determination. Regulatory minimum load is not the project design load.

https://www.access-board.gov/ada/vehicles/subpart-d/

C3B10 | BraunAbility - NCL1000 Century applicationguide,3June2026

Accessed 2026-10-17

NCL1000IB3754HB-2 retained row: 1,000 lb rating, 385 lb mass, 37 × 54 in platform, 50.96 in overall width, 45.85 in minimum doorway, 18 in threshold depth and 48 in travel.

Limit: Exact stowed CAD, rail application and mounting/turning fit unresolved. Analyst threshold rectangle is not the actual lift solid.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

C3B11 | Adaptive Engineering - Mobilift TX portable train lift

Accessed 2026-10-17

Rail station lift with manual operation, 600 lb rating, approximate 268 lb mass and floating bridge.

Limit: 600 lb is less than the 295 kg screening load. Factory proof test does not increase operating capacity. Terrain and every configured interface remain unverified.

https://adaptivelifts.com/tx.php

C3B12 | Garaventa - Genesis STAAGE portable lift

Accessed 2026-10-17

STAAGE 44: 750 lb, 44 in travel, 120 VAC; platform 36 × 52.75 in with lower gate or 36 × 54.5 in without; footprint 68.875 × 57.75 in; manual emergency lowering.

Limit: Stage application does not prove rail rescue fit. Rating fails 440/445/590 kg cases. Outdoor ground conditions, bridging, power and exact approval unresolved.

https://www.garaventalift.com/en/products/portable-lift-evacuation-chair/genesis-staage

C3B13 | Konvekta - Rail HVAC overview08/2021,7403

Accessed 2026-10-17

HVAC 7403 retained dimensions 3,100 × 1,600 × 415 mm without fasteners; mass 520 kg.

Limit: Geometry screen only. Roof attachments/CG, actual ambient performance, present availability and electrical compatibility unresolved.

https://www.konvekta.de/fileadmin/user_upload/docs/datenblaetter/schiene/Produktuebersicht_Schiene_dt-engl_eBook_WEB_0821.pdf

C3C01 | HydroAct hydraulic calipers

Accessed 2026-10-17

HSC1P10N is spring actuated, manufacturer brake force per disc20 kN, two hydraulic release circuits, maximum disc width45 mm, approximately15 kg. Force convention and release pressure not provided.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://rail.knorr-bremse.com/media/2000_products/2900d_product_broschures/p-1236_hydroact.pdf

C3C02 | EHM electro-hydraulic module

Accessed 2026-10-17

Independent electric/hydraulic activation of two bogies;24 VDC, train control wire48 VDC; CAN/MVB/Ethernet;1100x510x500 mm; approximately90 kg; proportional per bogie;IP65; permissible system pressure160 bar.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.hanning-kahl.com/products/hydraulic-brake-systems/electro-hydraulic-module.html

C3C03 | ECU series controller

Accessed 2026-10-17

Advertises replacement of HEYxPxG and HEY-C from2028; nominal24-110 VDC; conventional hardwired and CAN/MVB/Ethernet interfaces. Configured unit dimensions, current demand and project safety approval not supplied.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.hanning-kahl.com/products/brakemaster/control-unit-of-the-ecu-series.html

C3C04 | HCU hydraulic unit

Accessed 2026-10-17

Modular manifold, configuration-dependent functions, filtration and externally adjustable/sealable pressure relief valves. No project pressure setting, flow or port assignment established.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.hanning-kahl.com/products/brakemaster/hydraulic-unit-of-the-hcu-series.html

C3C05 | Auxiliary release devices

Accessed 2026-10-17

Auxiliary devices pressurize the second spring-actuator hydraulic circuit independently of service release. HP is a hand-pump family; HZY NL and HZY NL/HP include powered variants. This is a releasing function.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.hanning-kahl.com/products/hydraulic-brake-systems/auxiliary-release-devices.html

C3C06 | Electronic brake controllers

Accessed 2026-10-17

Project-specific braking matrix, analogue/digital and bus interfaces; optional wheel-slide functions. Prose describes certain SIL2 functions while table lists HEY-C SIL3 and HEY SIL2. Controlled certificate scope is unresolved; no system certification inferred.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.hanning-kahl.com/products/hydraulic-brake-systems/brake-electronic.html

C3C07 | People Mover semi-permanent coupler

Accessed 2026-10-17

Double-acting damper, pivot anchor and cable support documented. Configured dimensions, allowable motions, force/energy/fatigue ratings not published on inspected page.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.dellner.com/products/semi-permanent-couplers/semi-permanent-coupler-people-mover

C3C08 | Railway couplers

Accessed 2026-10-17

430/530 family advertised up to300 kN compression and tension. This family maximum does not rate a specific configured drawgear or its frame attachment; controlled configuration remains required.

Limit: Published documentation reviewed; no quote, stock, lead time or application acceptance obtained

https://www.voith.com/corp-en/connection-components-couplings/railway-couplers.html

C3F01 | AWS D15.1/D15.1M:2025 - railroad welding specification

Accessed 2026-10-17

Publisher identifies manufacture/maintenance of railcars, locomotives and components intended for North American railroad service; includes base metals thinner than 1/8 in. Errata link is published.

Limit: Public scope/product page read, not purchased complete standard; engineer must select applicability, edition/errata and project requirements.

https://pubs.aws.org/p/2260/d151d151m2025-railroad-welding-specification-for-cars-and-locomotives

C3F02 | AWS D1.1/D1.1M:2025-AMD1 - structural steel welding

Accessed 2026-10-17

AWS current page identifies steel welding procedure/welder qualification, fabrication, inspection and acceptance topics.

Limit: Candidate supplemental/contractual basis only; not automatically the railroad vehicle code or a Nevada legal requirement.

https://www.aws.org/standards-and-publications/codes-and-standards/d1-1/

C3F03 | AWS D1.2/D1.2M:2026 - structural aluminum welding

Accessed 2026-10-17

AWS current edition supersedes 2014 and covers welded aluminum structural requirements, qualification, fabrication and inspection.

Limit: Complete standard not reviewed; exact adopted edition and applicability belong to responsible designer.

https://www.aws.org/standards-and-publications/codes-and-standards/d1-2/

C3F04 | EN 15085 rail-vehicle welding certificate register

Accessed 2026-10-17

Register operated by SLV Halle presents certificates issued by accredited manufacturer certification bodies recognized by ECWRV.

Limit: A supplier qualification screening route, not evidence of any named supplier certificate or Nevada project compliance.

https://en15085.joincert.eu/?Sprache=2

C3F05 | TWI - distortion control by fabrication techniques

Accessed 2026-10-17

Tack layout, qualified procedures/personnel for incorporated tacks, balanced welding and planned sequences influence distortion.

Limit: General technical guidance; does not prescribe this frame weld sizes, heat input, tack spacing, preheat or correction limits.

https://www.twi-global.com/technical-knowledge/job-knowledge/distortion-control-prevention-by-fabrication-techniques-036

C3F06 | NIST policy on metrological traceability

Accessed 2026-10-17

Measurement traceability requires a documented calibration chain and associated uncertainty; a label alone does not establish fit for purpose.

Limit: No NIST certification of this project or shop claimed; shop determines measurement uncertainty against selected tolerances.

https://www.nist.gov/calibrations/traceability

C3F07 | Sika - Sikaflex-252 product data sheet

Accessed 2026-10-17

Vehicle assembly adhesive; tests on actual substrates and conditions are required to establish adhesion/material compatibility.

Limit: Candidate only. Typical values are not design allowables and no panel joint is qualified by this report.

https://usa.sika.com/dam/dms/us01/o/sikaflex-252.pdf

C3F08 | Sika - bonding/sealing with one-component Sikaflex, December 2025

Accessed 2026-10-17

Manufacturer general process guidance addresses substrate preparation, adhesion and cure considerations; coating/substrate choices affect adhesion.

Limit: Apply current product-specific approved process; generic guide is not this vehicle structural bond specification.

https://usa.sika.com/dam/dms/us01/0/General-Guideline-Bonding-Sealing-1-Component-Sikaflex-US-12-25.pdf